# AIL722 / AIL7022: Reinforcement Learning — Assignment 4

Cooperative MAPF as a Dec-POMDP & Offline RL Decoupled Scaling

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## Problem 1: Decentralized Multi-Agent Path Finding

### Subtask 1.1: Dec-POMDP Formulation & Theory

**(a) Formal Definition ⟨I, S, {A_i}, P, R, {Ω_i}, O, γ, b_0⟩:**
• **Agents**: I = {0, 1, 2, 3, 4} (N = 5).
• **State Space S**: s = (p, f, t, a^{t-1}) where p ∈ FreeCells^5 represents distinct agent positions, f ∈ {0, 1}^5 indicates absorbing goal status, t ∈ {0..80} is the timestep, and a^{t-1} is the joint previous action.
• **Transitions P**: Each unfinished agent independently slips with probability ε = 0.1 to Stay. Intended moves c_i are processed through iterative conflict resolution until a fixed point is reached: (1) boundary/wall collisions, (2) vertex conflicts, and (3) swap conflicts cascade. Goals are absorbing.
• **Rewards R**: Individual reward r_i^t = 10·1[reaches goal] - 0.05·1[unfinished] - 1.0·1[blocked]. Centralized team reward adds R_team = +5 once when all agents finish. Shaped reward adds β(γΦ(p') - Φ(p)) with Φ(p) = -d_i(p) and β = 0.1.
• **Observations Ω_i & O**: o_i = (F_i^t, δ_i^t, t/T, a_i^{t-1}) where F_i is a 4-channel 5×5 local window (walls, other agents, other goals, own goal), δ_i = (g_i - p_i)/11, and a_i^{t-1} is the one-hot past action (108 dims total).
• **Discount & Belief**: γ = 0.99, horizon T = 80; b_0 is a point mass on the fixed starting positions.

**(b) State Space Upper Bound & Tabular Infeasibility:**
With 65 free cells, position combinations are P(65, 5) = 65 × 64 × 63 × 62 × 61 ≈ 9.91 × 10^8. Including horizon steps (81) and previous actions (5^5 = 3,125), |S| ≤ 2.51 × 10^{14}. Joint action space |A|^N = 5^5 = 3,125. A joint tabular Q-table requires > 7.8 × 10^{17} entries. Tabular methods scale as O(F^N · |A|^N), exponential in the number of agents N and polynomial of degree N in grid cells F, rendering tabular methods strictly intractable.

**(c) Potential-Based Shaping Proof:**
For F(s, s') = γΦ(s') - Φ(s), the return telescopes over an episode of length τ: Q'π(s, a) = E[∑ γ^k r_k + ∑ γ^k(γΦ(s_{k+1}) - Φ(s_k))] = Q^π(s, a) - Φ(s) + E[γ^{τ+1}Φ(s_{τ+1})]. When Φ(s_terminal) = 0 at absorbing terminal states, the boundary term vanishes, yielding Q'π(s, a) = Q^π(s, a) - Φ(s). Because Φ(s) is action-independent, argmax_a Q'π(s, a) = argmax_a Q^π(s, a), preserving optimal policies. *Assumption Satisfaction*: For successful terminations, Φ(g_i) = -d_i(g_i) = 0, exactly satisfying the condition. For episodes truncated by horizon T = 80, Φ(s_T) ≠ 0, introducing a mild discounted bias βγ^{80}Φ(s_T) (γ^{80} ≈ 0.447) that rewards ending closer to goals.

### Subtask 1.2: Environment Dynamics & Unit Tests

All 11 unit tests in tests.py passed successfully:
1. **Wall/Boundary**: Moving into walls/boundary maintains position and incurs -1.0 penalty.
2. **Vertex Conflict**: Two agents targeting the same cell are both blocked.
3. **Swap Conflict**: Mutual head-on cell exchange is blocked for both agents.
4. **Following Move**: An agent cleanly enters a cell vacated in the same step.
5. **Cascade Blocking**: Obstruction cascades iteratively along a chain of 3 agents to fixed point.
6. **Rotation**: Four agents simultaneously cycle in a 2×2 free-cell block without conflict.
7. **Finished Agents**: Finished agents remain stationary, receive 0 reward, and act as static obstacles.
8. **Full Slip**: With ε = 1.0, actions become Stay without collision penalties.
9. **Reward Values**: Verified goal (+10), step (-0.05), collision (-1.0), team bonus (+5), and shaping identity.
10. **Determinism**: Identical seeds and action sequences produce exact matching trajectories.
11. **Observation Equivalence**: FastEnv flattened vectorized array matches PettingZoo dictionary output.

## Subtasks 1.3 & 1.4: IQL vs. QMIX Comparison

**Architecture & Hyperparameters**: Shared utility network $Q_\theta(o_i, id_i, a)$ uses Conv(4→32, 3×3) → Conv(32→32, 3×3) → Flatten (800) → Concat [800 + δ(2) + t/T(1) + prev_a(5) + id(5) = 813] → MLP(256, 256) → 5 actions. Replay buffer: 100,000 transitions; batch size 128 (640 agent samples); lr = 5×10^{-4}; ε decayed from 1.0 → 0.05 over 300,000 steps; hard target update every 1,000 steps.
**Does Centralized Training Help?** Yes. While IQL agents act greedily and frequently deadlock in narrow 1-cell corridors, QMIX trains a monotonic mixing network conditioned on global state (walls, all positions, goals, finished flags). This allows the mixer to coordinate joint turn-taking.
**Shaping Impact**: Potential-based shaping provides dense gradient signal towards goals, substantially accelerating early learning for both algorithms. However, shaped IQL exhibits higher collision rates as agents rush toward goals without coordination.
**Monotonicity Limitation**: QMIX's constraint $\partial Q_{tot} / \partial Q_i \geq 0$ cannot represent non-monotonic payoffs where an agent must take an individually suboptimal exploratory detour or retreat to enable a teammate's passage.

## Subtask 1.5: Visualization & Failure Analysis

Evaluation failures on 100 seeds are classified in priority order: *Goal Blocking* > *Deadlock* > *Livelock* > *Timeout*.
• **Deadlock**: Agents meet head-on in 1-cell corridors and neither retreats. Agents need memory/recurrency to break symmetry.
• **Goal Blocking**: A finished agent permanently occupies its goal, obstructing an unfinished agent's shortest path. Agents require global awareness beyond the 5×5 local FOV to anticipate congested bottleneck goals.
• **Livelock & Timeout**: Agents cycle between adjacent cells without progress. Overcome via communication or recurrent memory.

# Problem 2: Offline Reinforcement Learning

## Subtask 2.1: Decoupled Value and Policy Learning

**Implementation & Protocol**: Implemented in Q2/train_decoupled.py across **TD3+BC**, **IQL+AWR**, and **CQL+AWR** on mujoco/hopper/medium-v0 and D4RL/kitchen/complete-v2. The 1,000,000 gradient step budget is split into 200,000 critic steps and 800,000 policy extraction steps. Trained critics are frozen ($\nabla_\theta Q = 0$) and reused across policy data fractions to eliminate redundant compute.
**Nested Splitting**: Random subsets of {10%, 50%, 100%} are generated using a deterministic permutation such that $D_{10\%} \subset D_{50\%} \subset D_{100\%}$, isolating sample size effects from variance.
**Analysis (Park et al., 2024 Comparison)**:
Across the 3×3 scaling matrices, varying the *critic data fraction* along columns shows that value functions reach effective ranking accuracy even with only 10%–50% of critic transitions. Conversely, varying the *policy data fraction* along rows shows that policy extraction degrades sharply with reduced policy data. This directly corroborates the findings of Park et al.: **policy extraction is the primary bottleneck in offline RL**, not value learning. Confounding factors include MuJoCo rollout evaluation variance and step-budget allocations between critic and actor.

## Subtask 2.2: Implicit Q-Learning (IQL) from Scratch

**From-Scratch Implementation**: Implemented in Q2/train_iql_offline.py without off-the-shelf RL packages:
• **Value Network V($\psi$)**: Updated using asymmetric expectile regression $L_V(\psi) = E[|\tau - 1(Q_{target} - V < 0)|(Q_{target} - V)^2]$ with $\tau = 0.7$.
• **Twin Q-Networks Q($\theta$)**: Clipped double Q-learning targeting next-state value targets $y = r + \gamma(1 - d)V(s')$. Polyak target update $\tau_{soft} = 0.005$.
• **Advantage-Weighted Policy Extraction**: Diagonal Gaussian actor trained on exponentiated advantages $\exp((Q_{min} - V)/\beta) \nabla_\theta \log \pi(a|s)$ with $\beta = 3.0$, clipped at 100.0 for stability.
• **Evaluation & Visualization**: Evaluated over 5 independent seeds (0..4). Saved rollout animations as iql_seed{0..4}.gif.

| Hyperparameter | Value | Description |
|---|---|---|
| Expectile (τ) | 0.7 | Asymmetric value regression weight |
| Temperature (β) | 3.0 | Advantage exponentiation scale |
| Discount (γ) | 0.99 | Bellman discount factor |
| Target Update Rate (τ_soft) | 0.005 | Polyak target smoothing coefficient |
| Batch Size | 256 | Transitions per gradient update |
| Actor / Critic / Value LR | 3 × 10^{-4} | Adam optimizer learning rates |
| Actor LR Schedule | Cosine Annealing | Decays learning rate to 0 over 1M steps |
| Hidden Architecture | 256 × 256 | 2-layer MLP with ReLU activations |

## Deliverables & Reproducibility Summary

• **Submitted Scripts**: mapf_env.py, tests.py, train_iql.py, train_ctde.py, Q2/train_decoupled.py, Q2/train_iql_offline.py.
• **Visual Artifacts**: env_visualization.png, final_policy.gif, Q2/results/iql_seed{0..4}.gif.
• **Checkpoints & Logs**: Persisted in checkpoints/ and results/ directories for verification.